

Mingxuan Jin

PhD Researcher | Software Development and Interactive Systems

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EDUCATION

Cardiff University - PhD, Computer Science & Informatics School of Computational and Mathematical Sciences; expected completion Apr 2028	Apr 2024 - present
University of Nottingham Master's, Human-Computer Interaction (HCI) (Distinction)	Sep 2021 - Sep 2022
Beijing University of Technology & University College Dublin Bachelor's, Software Engineering	Sep 2016 - Jul 2020

INDUSTRY EXPERIENCE

Matrixtime Robotics Co., Ltd. <i>Deep Learning Technology Engineer Intern Shanghai, China</i> - Built a Python/PyTorch computer vision pipeline for automotive defect inspection, improving mAP by 15% and reducing false rejections from 8% to below 2%. - Deployed models with TensorRT FP16, reducing end-to-end latency per 1080p image from 65 ms to 18 ms for production-line use.	Mar 2023 - Jul 2023
China CITIC Bank <i>Banking Intern Client Services and Data Analysis</i> Automated banking-data cleaning and reporting with Python/pandas and Excel VBA, improving weekly reporting efficiency by around 30%. Flagged large related-party transfers in tens of thousands of transactions.	Jul 2018 - Aug 2018

SELECTED RESEARCH AND ENGINEERING

AR Walking System with IMU Integration and AI Feedback Validation Developer Magic Leap 2, Python <i>ACM Conference on Human Factors in Computing Systems 2027 (under review)</i> M. Zhou, Mingxuan Jin, K. Stawarz, X. Peng, A. Denisova, H. Egami, T. Wen, Y. Gu, D. J. Finnegan, S. C. Barathi. - Built a Python IMU-data pipeline and Random Forest movement classifier for lower-limb exercises. Implemented distance-based AR guide behaviour. - Integrated a language-model API with structured inputs. Validated feedback against fixed progression decisions and implemented deterministic fallback responses for inconsistent outputs.	Mar 2026 - Sep 2026
VR Cycling System with Gesture Input and Synchronised Logging Lead developer and evaluator Unity/C# <i>ACM Conference on Human Factors in Computing Systems 2027 (under review)</i> Mingxuan Jin, M. Zhou, X. Peng, D. J. Finnegan, L. D. Turner, S. C. Barathi. Integrated exercise-bike input with gesture-based dialogue. Implemented four controlled event sequences and synchronised power and event logging; led evaluation in studies with 16 and 68 children.	Mar 2026 - Sep 2026
Configurable VR Cycling Platform with Branching Dialogue Lead developer and evaluator Unity/C# <i>International Journal of Human-Computer Studies (Accepted for publication)</i> M. Zhou, Mingxuan Jin (co-first author), X. Peng, K. Stawarz, A. Denisova, J. Huang, S. C. Barathi. Built bike-controlled navigation, gesture-based branching dialogue and six experimental configurations. Led deployment and evaluation in two studies ($n = 15$ and $n = 36$).	May 2025 - Sep 2025

SELECTED SOFTWARE PROJECTS

NLP Review Classification and Editing Web App <i>Python, JavaScript, HTML/CSS</i> Combined Naive Bayes-derived aspect features with an SVM rating classifier. Built rule-based editing suggestions and a web interface, then iterated through usability testing.	May 2022 - Sep 2022
Bot Detection Web App with Model Explanations <i>Python, Django, JavaScript</i> Built a Django web app with a 12-feature Random Forest classifier, Bayesian hyperparameter optimisation and visual and natural-language explanations derived from decision trees.	Feb 2020 - Jun 2020

CORE SKILLS

Languages: Python, C#, Java, JavaScript, SQL, R
Tools: Unity, Django, PyTorch, TensorRT, pandas, MySQL, Git/GitHub
Systems: Bluetooth/IMU integration, API integration, synchronised event logging
Machine Learning: Computer vision, Random Forest, SVM